Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
3 kilometers = _____ meters

A. 0.3
B. 30
C. 300
D. 3,000

2. Which numbers are equivalent to $3\frac{1}{3}$? Circle all that apply.

A. 3.13
B. $3.\overline{3}$
C. $\frac{10}{3}$
D. $3\frac{2}{6}$
E. 3.3
F. $\frac{13}{3}$

3. Eighteen of 24 dogs at an animal shelter were adopted this month. What is the ratio of dogs that were adopted to dogs that were not adopted?

A. 3 : 4
B. 3 : 1
C. 1 : 3
D. 3 : 7

4. A class has a boys to girls ratio of 3 : 5. What is the ratio of girls to total students in the class?

A. 5 : 3
B. 3 : 8
C. 5 : 8
D. 8 : 5

5. Maya biked 27 miles in 1.5 hours. What was her speed in miles per hour?

A. 16 miles per hour
B. 18 miles per hour
C. 18.5 miles per hour
D. 40.5 miles per hour

6. Which rate should be used to convert $\frac{48 \text{ inches}}{1 \text{ minute}}$ to $\frac{\text{feet}}{\text{minute}}$?

A. $\frac{12 \text{ inches}}{1 \text{ foot}}$
B. $\frac{1 \text{ foot}}{12 \text{ inches}}$
C. $\frac{60 \text{ seconds}}{1 \text{ minute}}$
D. $\frac{1 \text{ minute}}{60 \text{ seconds}}$

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to walking 2,400 meters per hour.

7a. 40 meters per minute	**YES**	**NO**
7b. 144,000 meters per minute	**YES**	**NO**
7c. 57,600 meters per day	**YES**	**NO**
7d. 40 meters per second	**YES**	**NO**

8. What is the unit rate for $\frac{\frac{3}{4} \text{ mile}}{6 \text{ minutes}}$?

A. $4\frac{1}{2}$ miles per minute
B. 8 miles per minute
C. $\frac{1}{8}$ mile per minute
D. $\frac{3}{4}$ mile per minute

9. Ari skated $2\frac{1}{4}$ miles in $1\frac{1}{2}$ hours. Assume he skated the entire time at the same rate. What was his speed in miles per hour?

A. $\frac{2}{3}$ mile per hour
B. $1\frac{1}{2}$ miles per hour
C. $3\frac{3}{8}$ miles per hour
D. $1\frac{1}{6}$ miles per hour

Form B

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Part II – Constructed Response

Complete each conversion.

1. 9 feet = _____ inches

2. 3 meters = _____ centimeters

3. 40 ounces = _____ pounds

4. 18 inches = _____ feet

Convert each fraction to a decimal.

5. $\frac{3}{8}$

6. $\frac{5}{9}$

Convert each decimal to a fraction.

7. 0.6

8. 0.35

9. Simplify the ratio 16 : 24.

10. Fifteen of 20 students in the school band practiced over the weekend. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers): Explanation:	Ratio (numbers): Explanation:	Ratio (numbers): Explanation:

Find the unit rate.

11. $\frac{96 \text{ kilometers}}{4 \text{ hours}}$

12. $\frac{\frac{5}{8} \text{ meter}}{\frac{3}{4} \text{ second}}$

13. Rosa spent $54.00 on six tickets to a basketball game. Dev spent $44 on four tickets. Which person paid less per ticket? Show all work necessary to justify your answer.

14. A cyclist rides at a rate of 15 miles per hour. Write this rate as feet per hour. Show all work necessary to justify your answer.

15. A ferry travels at a rate of 18 kilometers per hour. Write this rate as meters per minute. Show all work necessary to justify your answer.

16. It took $\frac{3}{4}$ hour to wash 6 cars at a fundraiser. Assume each car took the same amount of time. What was the rate in hours per car? Nora answered this question, but made a mistake. Explain Nora's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{\frac{3}{4}\text{ hour}}{6\text{ cars}}$

$\frac{\frac{3}{4}\text{ hour}}{6\text{ cars}} = \frac{3}{4} \div \frac{1}{6}$

$= \frac{3}{4} \cdot \frac{6}{1}$

$= \frac{18}{4}$

$= 4\frac{1}{2}$ hours per car

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Part III – Problem Solving

There are 10 girls and 15 boys on the swim team. They swim laps in a swim-a-thon to raise money for new team suits. Each girl raised $6.00 per lap and swam at a rate of 12 laps per hour. Each boy raised $4.00 per lap and swam at a rate of 2 laps every 15 minutes. What is the ratio of the amount of money the girls raised in one hour to the amount of money the boys raised in one hour?